

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Chemistry Semester-III Mid-Semester Examination 2025-26 (NEP)

CHEMISTRY**Paper Code: CHEMJ33 : Physical Chemistry-I**

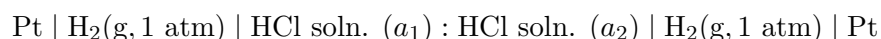
Date: 04.11.2025

Time: 1 Hour

Maximum Marks: 20

Instructions: Answer all the questions. Figures in the right-hand margin indicate marks.

1. Define molar conductance and equivalent conductance of an electrolyte. Why does the conductivity of a solution decrease with dilution? [2]
2. Calculate the degree of dissociation (α) of acetic acid if its molar conductivity (Λ_m) is $39.05 \text{ S cm}^2 \text{ mol}^{-1}$. Given: $\lambda^\circ(\text{H}^+) = 349.6 \text{ S cm}^2 \text{ mol}^{-1}$ and $\lambda^\circ(\text{CH}_3\text{COO}^-) = 40.9 \text{ S cm}^2 \text{ mol}^{-1}$. [2]
3. Define transference (transport) number of ions. Write the relation between transference number and ionic mobility. [2]
4. Plot and explain a graph for the conductometric titration of weak acid by strong base. [2]
5. A moving boundary experiment is conducted to measure the transference number of the Li^+ ion in a 0.01 mol L^{-1} LiCl solution. In a tube having a cross-sectional area of 0.125 cm^2 , the boundary moves 7.3 cm in 1490 sec using a current of $1.8 \times 10^{-3} \text{ A}$. Calculate the transference numbers of the cation and anion. [2]
6. Define cell potential. How can you determine the standard electrode potential of an electrode? [2]
7. What is liquid junction potential? How does a salt bridge minimise the liquid junction potential of a cell? [2]
8. For the concentration cell:



write the various processes at the two electrodes and at the liquid junction. [3]

9. A 50 mL 0.05 M solution of Fe(II) is titrated with 0.05 M solution of Ce(IV) in the presence of dilute H_2SO_4 at 25°C by coupling with saturated calomel electrode. Draw the potentiometric curve (E_{cell} vs. volume of Ce(IV)) and calculate the equivalence point potential. [Given, $E^\circ_{\text{Fe}^{3+}/\text{Fe}^{2+}} = 0.75 \text{ V}$ and $E^\circ_{\text{Ce}^{4+}/\text{Ce}^{3+}} = 1.45 \text{ V}$ at 25°C] [3]
